



Lab Report

Submitted By
Submitted to
Roll No
Date

Ijlal Ahmad
Mr. Aqib Adeel
FL-21565
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Lab 01

```
#include<iostream>
using namespace std;
int main(){

double salary, tax, remsal,a,b;

cout<<"enter salary";

cin>>salary;

if(salary<=10000){

    tax=salary*0.05;

    remsal=salary-tax;

    cout<<"Your salary after tax"<<remsal;

}

else if(salary>10000){

    a=(salary-10000)*0.1;

    b=10000*0.05;

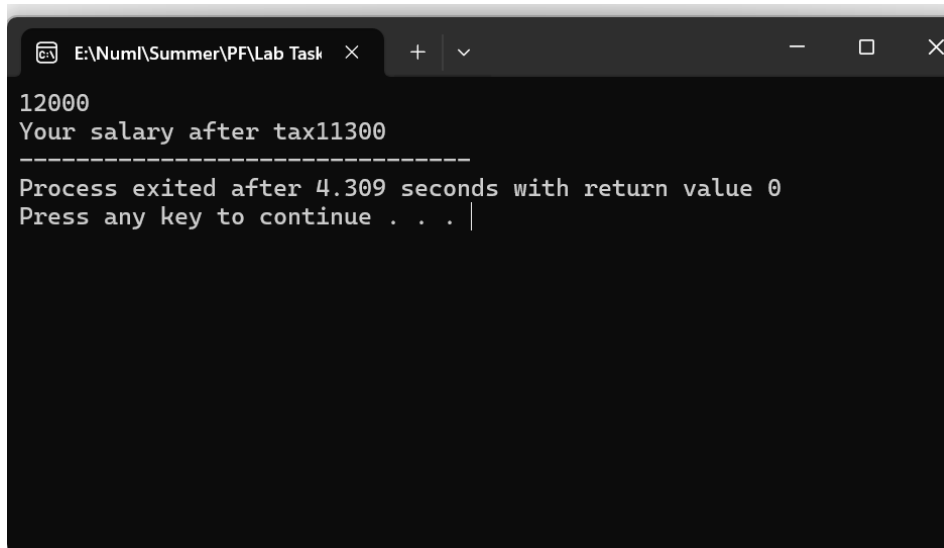
    tax=a+b;

    remsal=salary-tax;

    cout<<"Your salary after tax"<<remsal;

}

}
```

A screenshot of a Windows command prompt window. The title bar shows the file path 'E:\Num1\Summer\PF\Lab Task' and standard window controls. The output text is as follows:

```
12000
Your salary after tax11300
-----
Process exited after 4.309 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){

    int unit, bill, a, b, c, d, e;

    cout<<"enter No of units:";

    cin>>unit;

    if(unit<=100){

        bill=unit*5;

        cout<<"your bill is:"<<bill;

    }

    else if(unit>100 && unit<=200){

        a=unit-100;

        a=a*10;

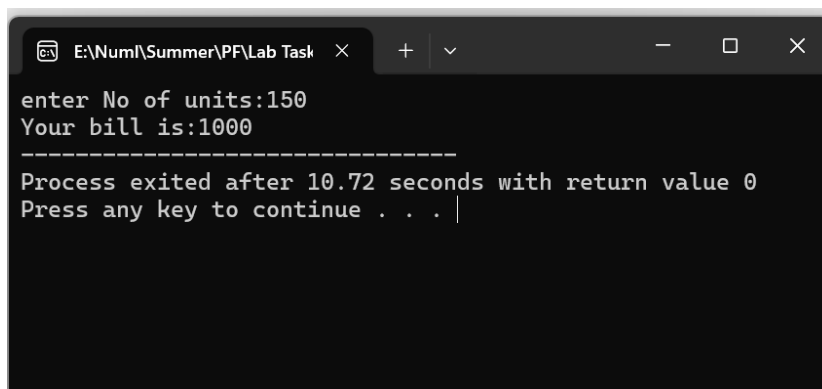
        b=100*5;

        cout<<"Your bill is:"<<a+b;
```

```

    }
    else{
        c=unit-200;
        c=c*15;
        d=100*10;
        e=100*5;
        cout<<"Your bill is:"<<c+d+e;
    }
}

```



```

E:\Num\Summer\PF\Lab Task
enter No of units:150
Your bill is:1000
-----
Process exited after 10.72 seconds with return value 0
Press any key to continue . . . |

```

```

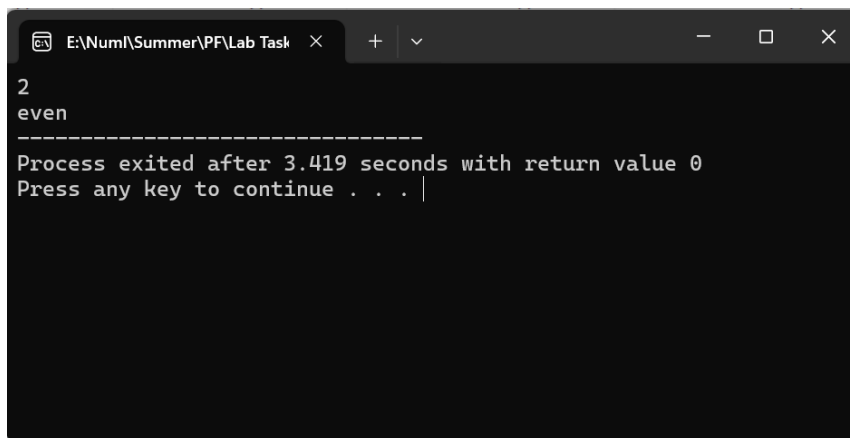
#include<iostream>
using namespace std;
int main(){
    int a;
    cin>>a;
    switch(a%2){
        case 0:
            cout<<"even";
            break;

```

```

        case 1:
            cout<<"odd";
            break;
    }
}

```



```

E:\Num\Summer\PF\Lab Task
2
even
-----
Process exited after 3.419 seconds with return value 0
Press any key to continue . . . |

```

```

#include<iostream>
using namespace std;
int main(){
    int a,b;
    char c;
    cin>>a>>b;
    cout<<"enter the sign";
    cin>>c;
    switch(c){
        case 'p':

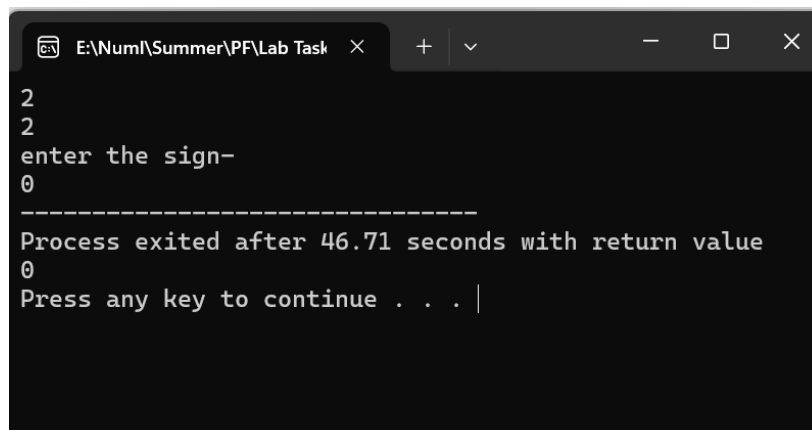
```

```

        case '+':
            cout<<a+b;
            break;

        case 'm':
        case '-':
            cout<<a-b;
            break;
    }
}

```



The screenshot shows a Windows command prompt window with the title bar "E:\Num1\Summer\PF\Lab Task". The window contains the following text:

```

2
2
enter the sign-
0
-----
Process exited after 46.71 seconds with return value
0
Press any key to continue . . . |

```

```

#include<iostream>
using namespace std;
int main(){
    int m;
    cin>>m;

```

```
if(m<0 || m>100){
    cout<<"invalid";
}
else{
    switch(m/10){
        case 10:
        case 9:
        case 8:
            cout<<"Grade a";
            break;

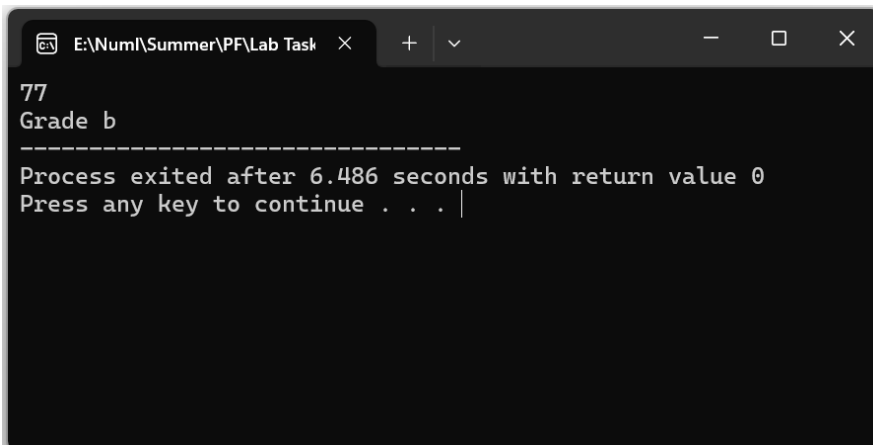
        case 7:
            cout<<"Grade b";
            break;

        case 6:
            cout<<"Grade c";
            break;

        case 5:
            cout<<"Grade d";
            break;

        case 4:
        case 3:
        case 2 :
        case 1:
            cout<<"fail";
            break;
```

}}



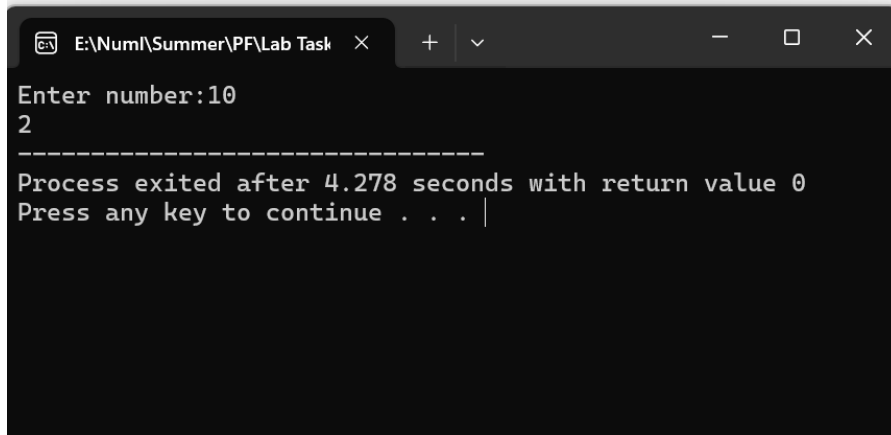
A screenshot of a Windows command prompt window. The title bar shows the file path "E:\Num1\Summer\PF\Lab Task" and standard window controls. The command prompt displays the output of a C++ program: the number "77", the text "Grade b", a separator line of dashes, and a message "Process exited after 6.486 seconds with return value 0". It also shows the prompt "Press any key to continue . . . |" with a cursor.

```
E:\Num1\Summer\PF\Lab Task
77
Grade b
-----
Process exited after 6.486 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){
    int count=0;
    int a;
    cout<<"Enter number:";
    cin>>a;
    while(a!=0){
        count=count+1;
        a=a/10;
    }
    cout<<count;
}
```

```
E:\Numl\Summer\PF\Lab Task
Enter number:10
2
-----
Process exited after 4.278 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){

int a;

cout<<"Enter a number between 1 and 10:";

cin>>a;

while(a<1 || a>10){

    cout<<"enter the number in a limit\n";

    cout<<"enter again";

    cin>>a;

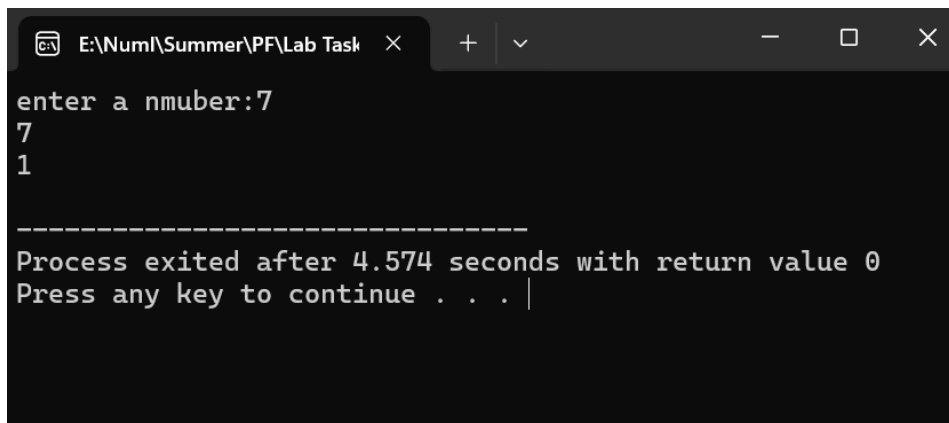
}

cout<<"you enter:"<<a;

}
```

```
E:\Num1\Summer\PF\Lab Task  ×  +  ∨  
Enter a number between 1 and 10:12  
enter the number in a limit  
enter again9  
you enter:9  
-----  
Process exited after 11.72 seconds with return value 0  
Press any key to continue . . . |
```

```
#include<iostream>  
  
using namespace std;  
  
int main(){  
    int a, num;  
  
    cout<<"enter a nmuber:";  
  
    cin>>num;  
  
    a=num;  
  
    while(a>=1){  
        if(num%a==0){  
            cout<<a<<endl;  
        }  
        a=a-1;  
    }  
}
```



```
enter a nmuber:7
7
1

-----
Process exited after 4.574 seconds with return value 0
Press any key to continue . . . |
```

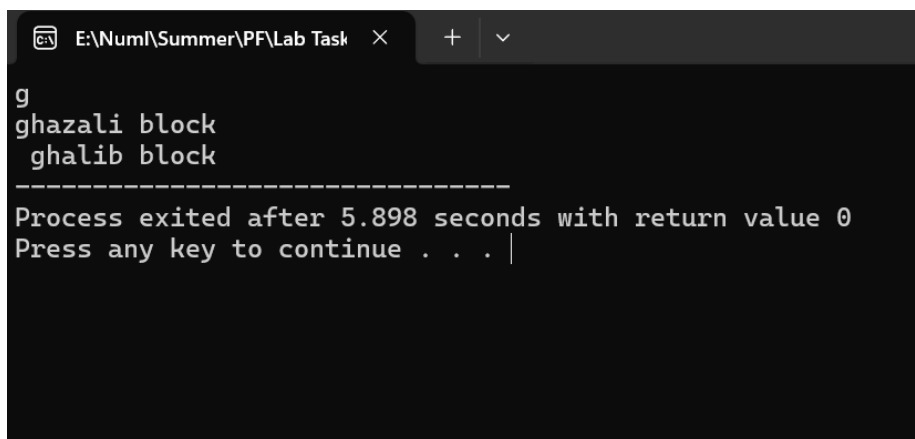
```
#include<iostream>
using namespace std;
int main(){
char a;
cin>>a;
switch(a){
    char i,g,r;
    case 'i':
        cout<<"iqbal block \n ibn-e-rushd block \n ibn-e-khaldun block";
        break;

    case 'g':
        cout<<"ghazali block \n ghalib block";
        break;
```

```
        case 'r':
            cout<<"rumi block";
            break;

        default:
            cout<<"invalid";
    }

}
```



```
E:\Numl\Summer\PF\Lab Task
g
ghazali block
ghalib block
-----
Process exited after 5.898 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){

int marks;
```

```
cout<<"Enter marks between 0 and 100:";
cin>>marks;
```

```
if(marks>=0 && marks<=39){
    cout<<"Grade F:";
}
```

```
else if(marks>=40 && marks<=54){
    cout<<"grade D";
}
```

```
else if(marks>=55 && marks<=69){
    cout<<"grade c";
}
```

```
else if(marks>=70 && marks<=84){
    cout<<"grade B";
}
```

```
else if(marks>=85 && marks<=100){
    cout<<"grade A";
}
```

```
else{
```

```
    cout<<"marks invalid Enter again";
}}
```

```
Enter marks between 0 and 100:67
grade c
-----
Process exited after 3.366 seconds with return value 0
Press any key to continue . . . |
```

Lab 02

```
#include<iostream>

using namespace std;

int main(){

    int n;

    cout<<"enter a number";

    cin>>n;


    int a,b,c;

    cin>>b;

    cin>>c;

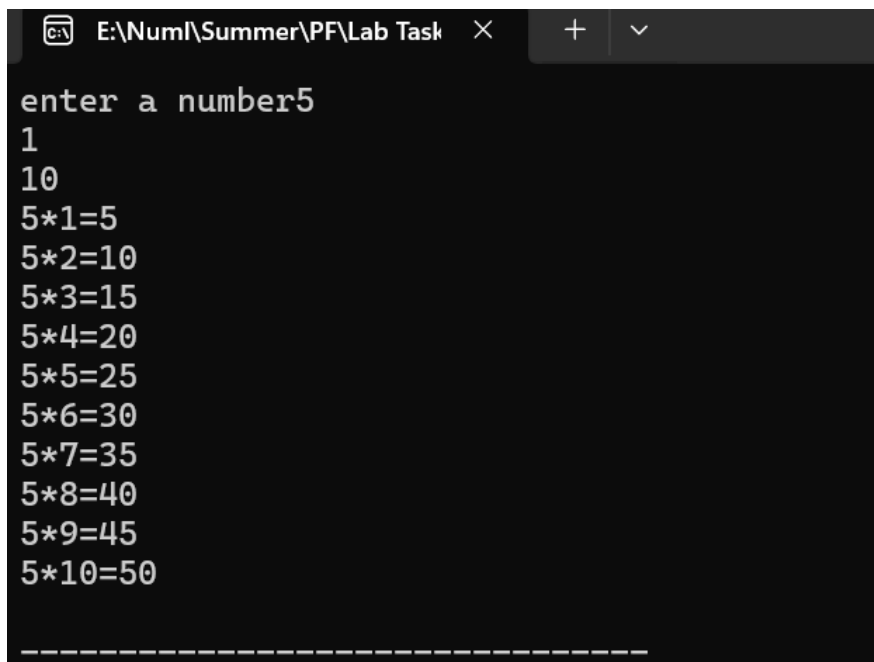

    a=b;

    while(a<=c){

        cout<<n<<"*"<<a<<"="<<n*a<<endl;

        a=a+1;

    }}
```



```
E:\Num1\Summer\PF\Lab Task × + v
enter a number5
1
10
5*1=5
5*2=10
5*3=15
5*4=20
5*5=25
5*6=30
5*7=35
5*8=40
5*9=45
5*10=50
-----
```

```
#include<iostream>

using namespace std;

int main(){

int num, sum=0;

cout<<"enter a number";

cin>>num;

while(num!=-1){

cin>>num;

sum=sum+num;

}

cout<<"the sum is:"<<sum;
```

```
}
```

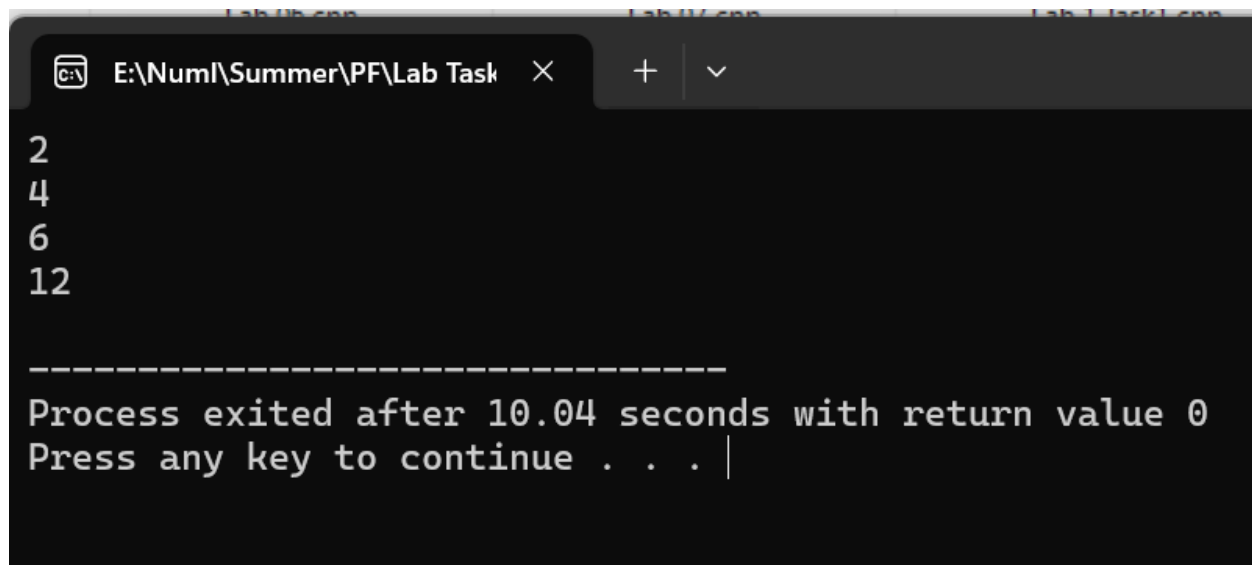
```
enter a number10
2
7
91
-1
the sum is:99
-----
Process exited after 14.31 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>
using namespace std;
int main(){
int fact=1, num;
cout<<"enter a number:";
cin>>num;
for(int i=1; i<=num; i++){
    fact=fact*i;
}
cout<<"factorial of "<<num<<"is"<<fact;
}
```



```
E:\Num1\Summer\PF\Lab Task  X  +  v
enter a number:5
factorial of 5is120
-----
Process exited after 3.953 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>
using namespace std;
int main(){
    int a, b, c;
    cin>>a>>b>>c;
    for(int i=1; i<=a*b*c; i++){
        if(i%a==0 && i%b==0 && i%c==0){
            cout<<i<<endl;
            break;
        }
    }
}
```



```
E:\Num1\Summer\PF\Lab Task  X  +  v
2
4
6
12

-----
Process exited after 10.04 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){
    int n, count;
    cout<<"enter number:";
    cin>>n;
    for(int a=1; a<=n; a++){
        if(n%a==0){
            cout<<a<<" ";

        }
    }
}
```

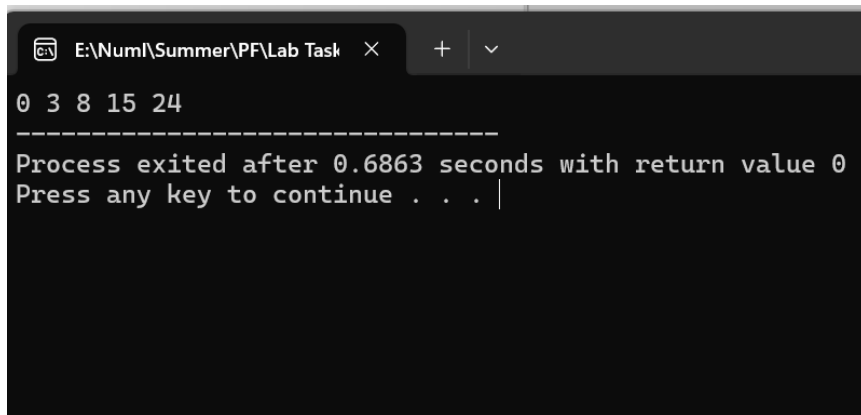
```
}
```

```
enter number:10
1 2 5 10
-----
Process exited after 2.013 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){
    int z;
    for(z=1; z<=5; z++){
        cout<<z*z-1<<" ";
    }
}
```



```
E:\Numl\Summer\PF\Lab Task  ×  +  ∨  
0 3 8 15 24  
-----  
Process exited after 0.6863 seconds with return value 0  
Press any key to continue . . . |
```

```
#include<iostream>  
  
using namespace std;  
  
int main(){  
    int i=10;  
    while(i>=5){  
        int j=5;  
        while(j<=10){  
            cout<<"Ijlal Ahmad\n";  
            j=j+2;  
        }  
  
        i=i-2;  
    }  
}
```

```
E:\Num1\Summer\PF\Lab Task  ×  +  ∨  
Ijlal Ahmad  
Ijlal Ahmad  
Ijlal Ahmad  
Ijlal Ahmad  
Ijlal Ahmad  
Ijlal Ahmad  
Ijlal Ahmad  
Ijlal Ahmad  
Ijlal Ahmad  
-----  
Process exited after 0.826 seconds with return value 0  
Press any key to continue . . . |
```

```
#include<iostream>  
  
using namespace std;  
  
int main(){  
  
int i=1;  
  
while(i<=3){  
  
int j=1;  
  
while(j<=3){  
  
cout<<"i="<<i<<" j="<<j<<endl;  
  
j=j+1;  
  
}  
  
i=i+1;  
  
}}
```

```
E:\Num1\Summer\PF\Lab Task  ×  +  v
i=1 j=1
i=1 j=2
i=1 j=3
i=2 j=1
i=2 j=2
i=2 j=3
i=3 j=1
i=3 j=2
i=3 j=3

-----
Process exited after 0.784 seconds with return code 0
Press any key to continue . . .
```

```
#include<iostream>

using namespace std;

int main(){

int a, b;

for(a=1; a<=5; a++){

for(b=1; b<=5; b++){

if(a%2==0){

cout<<"1";

}

else{

cout<<"0";

}

}

cout<<endl;

}}
```

```
E:\Num1\Summer\PF\Lab Task × + v
00000
11111
00000
11111
00000

-----
Process exited after 0.6987 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>
using namespace std;
int main(){
int a, b;
for(a=1; a<=6; a++){
for(b=1; b<=7; b++){
if(b%2==0){
cout<<"0";
}
else{
cout<<"1";
}
}
cout<<endl;
}}
```

```
E:\Numl\Summer\PF\Lab Task  X  +  v
1010101
1010101
1010101
1010101
1010101
1010101
1010101

-----
Process exited after 0.7084 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){

int a, b;

for(a=1; a<=4; a++){

for(b=1; b<=5; b++){

if(a==b){

cout<<"1";

}

else{

cout<<"0";

}

}

cout<<endl;

}}
```



```
E:\Num\Summer\PF\Lab Task  ×  +  ▾
10000
01000
00100
00010

-----
Process exited after 0.6794 seconds with return value 0
Press any key to continue . . . |
```

Lab 03

```
#include<iostream>

using namespace std;

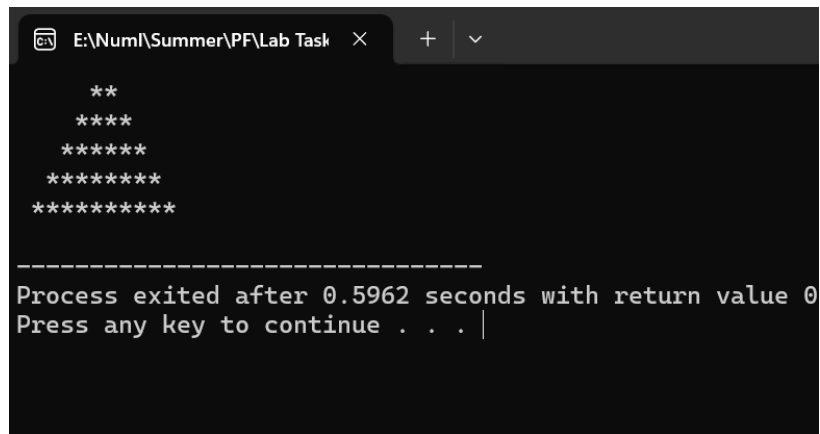
int main(){

    int a, b;

    for(a=1; a<=3; a++){
        for(b=1; b<=5; b++){
            if(a==1 || a==3 || b==1 || b==5){
                cout<<"1";
            }
            else{
                cout<<" ";
            }
        }
        cout<<endl;
    }
}
```

```
E:\Num1\Summer\PF\Lab Task  ×  +  ∨  
11111  
1  1  
11111  
  
-----  
Process exited after 0.7788 seconds with return  
Press any key to continue . . . |
```

```
#include<iostream>  
  
using namespace std;  
  
int main(){  
    int a, b, j;  
  
    for(a=1; a<=5; a++){  
  
        for(b=a; b<=5; b++){  
  
            cout<<" ";  
        }  
  
        for(j=1; j<=a*2; j++){  
            cout<<"*";  
        }  
        cout<<endl;  
    }  
}
```



```

**
****
*****
*****
*****

-----
Process exited after 0.5962 seconds with return value 0
Press any key to continue . . . |

```

```
#include<iostream>

using namespace std;

int main(){

    for(int a=5; a>=1; a--){

        for(int b=a; b<=5; b++){

            cout<<" ";

        }

        for(int j=1; j<=a*2-1; j++){

            cout<<"*";

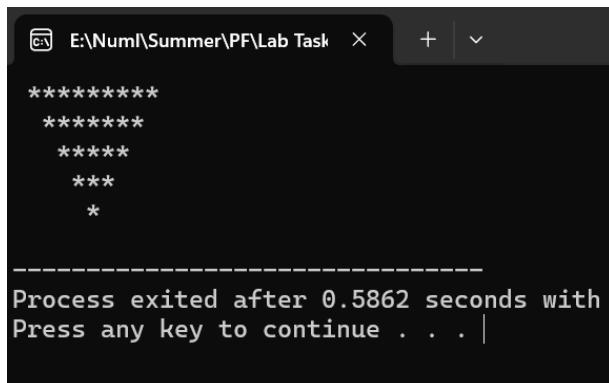
        }

        cout<<endl;

    }

}
```

}



```
E:\Numl\Summer\PF\Lab Task X + v
*****
*****
*****
***
*

-----
Process exited after 0.5862 seconds with
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){
    int size=5;
    int arr[size];

    for(int i=0; i<size; i++){
        cout<<"enter the element at index "<<i<<endl;
        cin>>arr[i];
    }

    for(int i=0; i<size; i++){
        cout<<"the element at index "<<i<<" is:"<<arr[i]<<endl;
    }
}
```

```
E:\Numl\Summer\PF\Lab Task  ×  +  ▾
enter the element at index 0
20
enter the element at index 1
34
enter the element at index 2
45
enter the element at index 3
87
enter the element at index 4
7
the element at index 0 is:20
the element at index 1 is:34
the element at index 2 is:45
the element at index 3 is:87
the element at index 4 is:7

-----
Process exited after 17.87 seconds with return value 0
```

```
#include<iostream>

using namespace std;

int main(){

    const int size=5;

    int sum=0, avrg;

    int fee[size];

    for(int i=0; i<size; i++){

        cout<<"Enter Fee";

        cin>>fee[i];

    }

    for(int i=0; i<size; i++){

        sum=sum+fee[i];

        avrg=sum/size;
```

```

}

cout<<sum<<endl;

cout<<avrg;

}

```

```

E:\Numl\Summer\PF\Lab Task
Enter Fee1000
Enter Fee2000
Enter Fee4000
Enter Fee3000
Enter Fee6000
16000
3200
-----
Process exited after 20.12 seconds with return v
Press any key to continue . . .

```

```

#include<iostream>

using namespace std;

int main(){

int n=10;

int count=0;

int guess;

cout<<"Enter a number between 1 & 20\n";

cin>>guess;

do{

```

```
if(guess==n)
{
cout<<"You guess the right number";
break;
}
```

```
else if(guess<n)
{
cout<<"too low ";
count++;
cout<<3-count<<" attempts left";
if(count<3){
cin>>guess;}
}
```

```
else if(guess>n)
{
cout<<"too high ";
count++;
    cout<<3-count<<" attempts left";
    if(count<3){
cin>>guess; }
}
}
```

```
while(count<3);
}
```

```
E:\Num1\Summer\PF\Lab Task  X  +  v
Enter a number between 1 & 20
12
too high 2 attempts left2
too low 1 attempts left10
You guess the right number
-----
Process exited after 15.1 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){

int a=5;

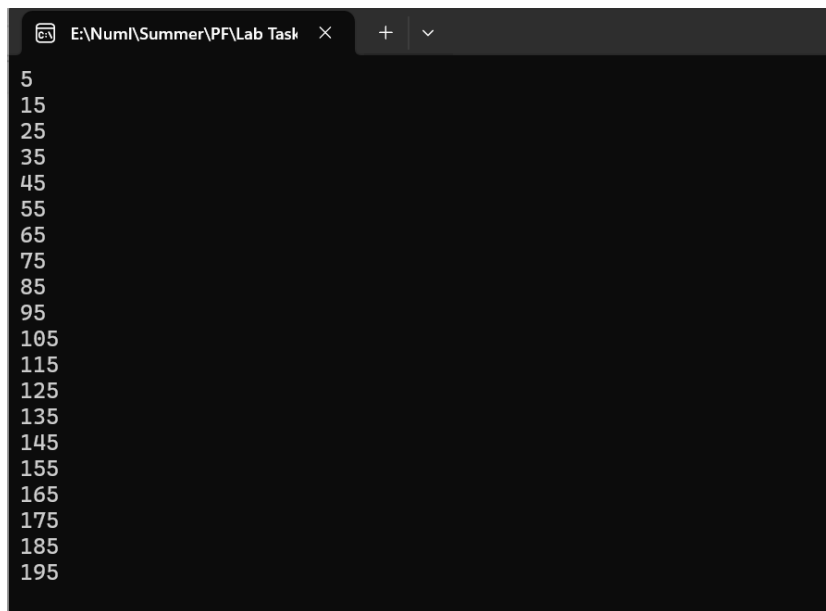
for(int i=1; i<=20; i++){

cout<<a<<endl;

a=a+10;

}

}
```


A screenshot of a terminal window with a dark background. The window title bar shows the file path "E:\Num1\Summer\PF\Lab Task" and standard window controls. The terminal displays a list of numbers from 5 to 195 in increments of 10, arranged in a single column.

```
5
15
25
35
45
55
65
75
85
95
105
115
125
135
145
155
165
175
185
195
```

```
#include<iostream>

using namespace std;

int main(){

int s=5 ,n;

float myarr[s];

for(int b=0; b<s; b++){

cout<<"Enter a anumber";

cin>>myarr[b];

}

cout<<"Enter a number to search";

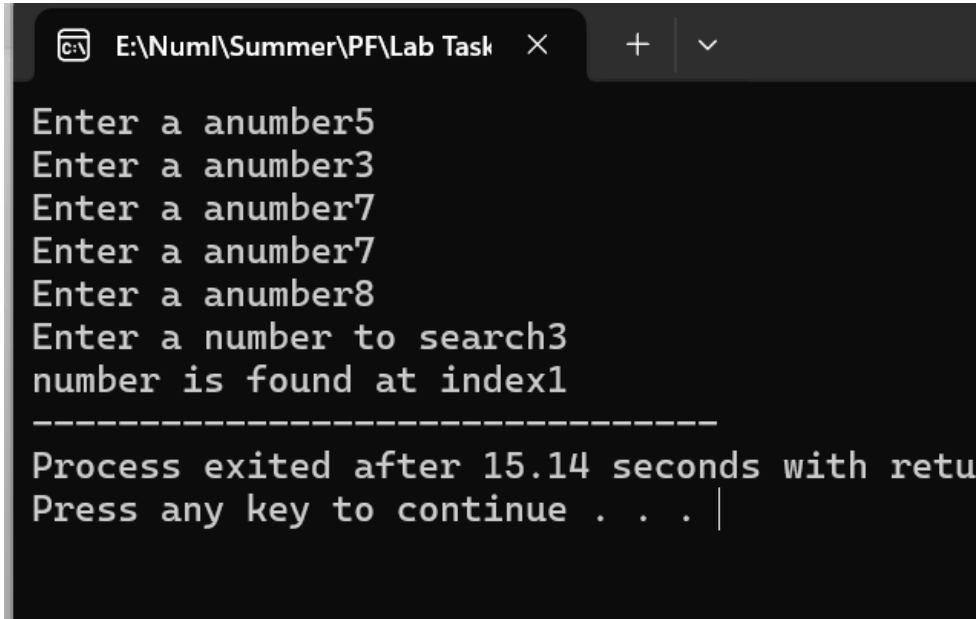
cin>>n;

for(int b=0; b<s; b++){
```

```

if(n==myarr[b]){
cout<<"number is found at index"<<b;
break;
}
}}

```



```

E:\Numl\Summer\PF\Lab Task
Enter a number5
Enter a number3
Enter a number7
Enter a number7
Enter a number8
Enter a number to search3
number is found at index1
-----
Process exited after 15.14 seconds with return code 0
Press any key to continue . . .

```

```

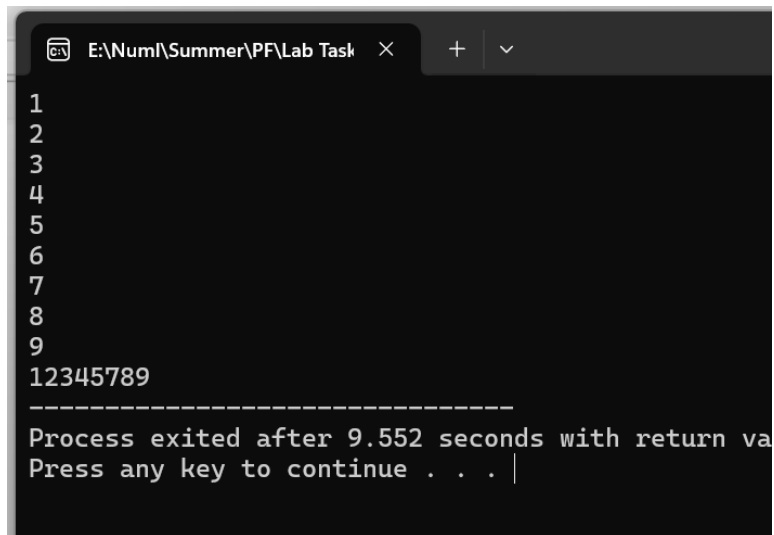
#include<iostream>
using namespace std;
int main(){
int abc[4][3];
cin>>abc[0][0];
cin>>abc[0][1];
cin>>abc[0][2];
cin>>abc[1][0];
cin>>abc[1][1];
cin>>abc[1][2];

```

```
cin>>abc[2][0];
cin>>abc[2][1];
cin>>abc[2][2];

cout<<abc[0][0];
cout<<abc[0][1];
cout<<abc[0][2];
cout<<abc[1][0];
cout<<abc[1][1];
cout<<abc[2][0];
cout<<abc[2][1];
cout<<abc[2][2];

}
```



```
E:\Numl\Summer\PF\Lab Task  X + v
1
2
3
4
5
6
7
8
9
12345789
-----
Process exited after 9.552 seconds with return va
Press any key to continue . . . |
```

Lab 05

```
#include<iostream>

using namespace std;

int main(){

    int num;

    const int rows=3, col=3;

    int arr[rows][col];


    for (int i=0; i<rows; i++){
        for (int j=0; j<col; j++){
            cin>>arr[i][j];

        }
    }

    for (int i=0; i<rows; i++){
        for (int j=0; j<col; j++){
            cout<<arr[i][j]<<"\t";
        }
        cout<<endl;
    }
}
```

```
g++  VIEW  PROJECT  EXECUTE  TOOLS  ASVIEW  WINDOW  FIELD
E:\Num1\Summer\PF\Lab Task  ×  +  ∨
1
3
5
7
9
8
2
4
6
1      3      5
7      9      8
2      4      6

-----
Process exited after 9.461 seconds with return value
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){

int num;

const int rows=4, col=3;

int mat[rows][col];

int sum=0;

for (int i=0; i<rows; i++){
for (int j=0; j<col; j++){
cin>>mat[i][j];
}
}
```

```
for (int i=0; i<rows; i++){
```

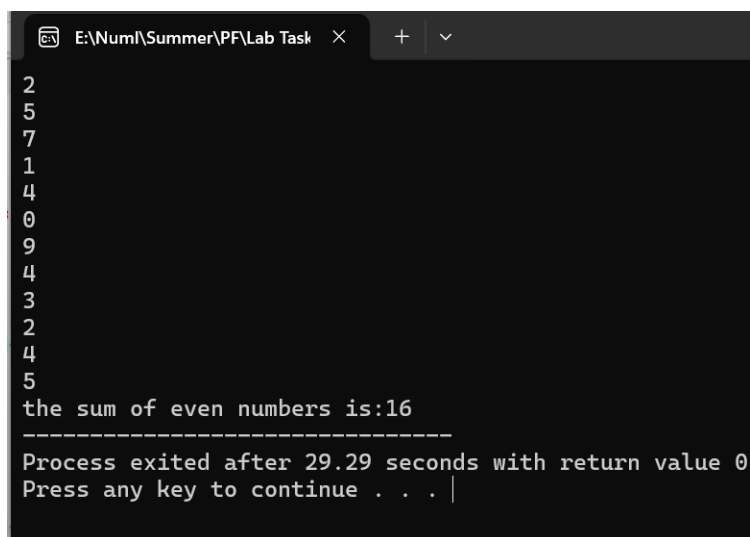
```

for (int j=0; j<col; j++){
    if(mat[i][j]%2==0){
        sum=sum+mat[i][j];
    }
}

cout<<"the sum of even numbers is:"<<sum;

}

```



```

E:\Numl\Summer\PF\Lab Task
2
5
7
1
4
0
9
4
3
2
4
5
the sum of even numbers is:16
-----
Process exited after 29.29 seconds with return value 0
Press any key to continue . . .

```

```

#include<iostream>

using namespace std;

int main(){

    int num;

    const int rows=2, col=2;

    int mat[rows][col];

    int found=0;

    for (int i=0; i<rows; i++){

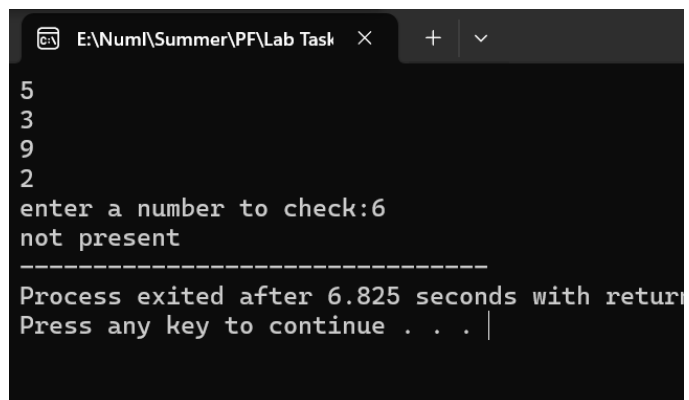
```

```

for (int j=0; j<col; j++){
cin>>mat[i][j];
}
}

cout<<"enter a number to check:";
cin>>num;
for (int i=0; i<rows; i++)
{
for (int j=0; j<col; j++)
{
if(num==mat[i][j])
{
cout<<" nummber present";
found=1;
}}}
if(found==0){
cout<<"not present";
}}

```



```

E:\Numl\Summer\PF\Lab Task
5
3
9
2
enter a number to check:6
not present
-----
Process exited after 6.825 seconds with return
Press any key to continue . . . |

```

```
#include<iostream>

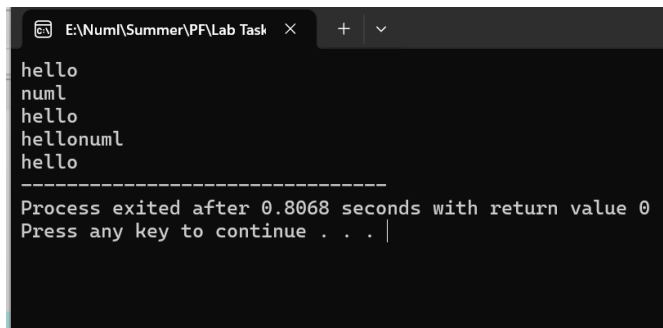
using namespace std;
```

```
void hello();
```

```
main(){
hello();
cout<<endl;
cout<<"numl\n";
hello();
cout<<endl;
hello();
cout<<"numl\n";
hello();

}
```

```
void hello(){
cout<<"hello";
}
```

A screenshot of a Windows command prompt window. The title bar shows the file path "E:\Numl\Summer\PF\Lab Task" and standard window controls. The command prompt displays the output of the program: "hello", "numl", "hello", "hellonuml", and "hello" on separate lines. Below the output, a separator line is followed by the text "Process exited after 0.8068 seconds with return value 0" and "Press any key to continue . . . |".

```
hello
numl
hello
hellonuml
hello
-----
Process exited after 0.8068 seconds with return value 0
Press any key to continue . . . |
```

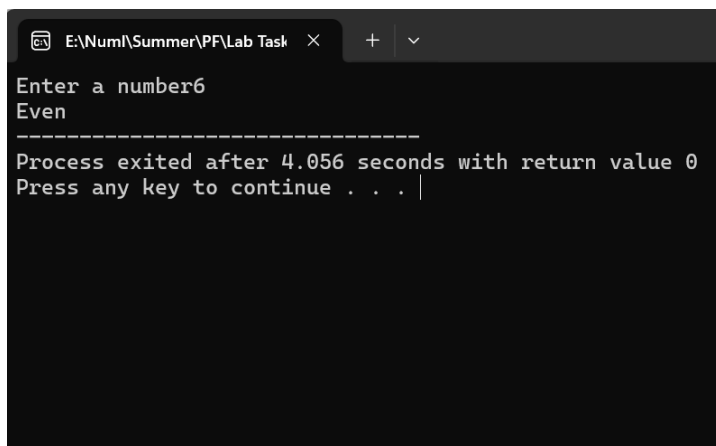
```
#include<iostream>
```



```
using namespace std;

void check_event(){
    int a;
    cout<<"Enter a number";
    cin>>a;
    if(a%2==0){
        cout<<"Even";
    }
    else{
        cout<<"odd";
    }
}
```

```
main(){
    check_event();
}
```



```
E:\Numl\Summer\PF\Lab Task  ×  +  ▾
Enter a number6
Even
-----
Process exited after 4.056 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

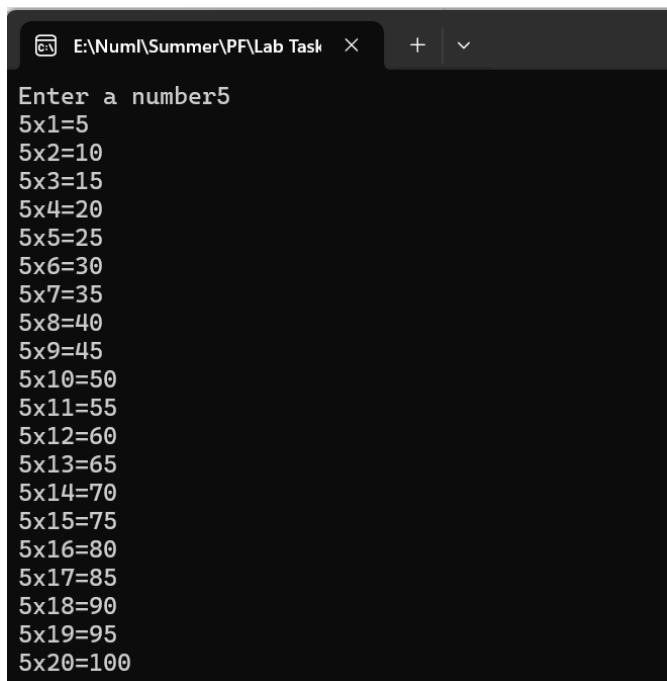
using namespace std;
```

```

void print_table(){
    int a, i;
    cout<<"Enter a number";
    cin>>a;
    for(int i=1; i<=20; i++){
        cout<<a<<"x"<<i<<"="<<a*i<<endl;
    }
}

main(){
    print_table();
}

```



```

E:\Num1\Summer\PF\Lab Task
Enter a number5
5x1=5
5x2=10
5x3=15
5x4=20
5x5=25
5x6=30
5x7=35
5x8=40
5x9=45
5x10=50
5x11=55
5x12=60
5x13=65
5x14=70
5x15=75
5x16=80
5x17=85
5x18=90
5x19=95
5x20=100

```

```

#include<iostream>
using namespace std;

```

```

void add(int, int, int);

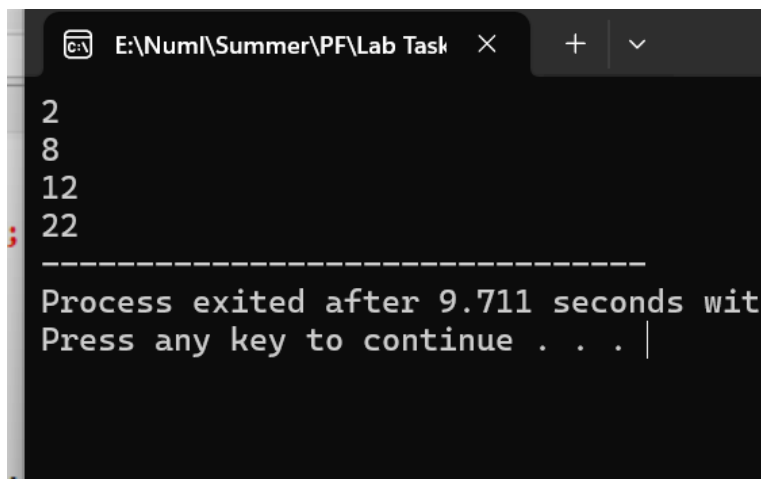
main(){
    int a,b,c;
    cin>>a>>b>>c;
    add(a,b,c);
}

```

```

void add(int f, int g, int h){
    //formal arguments
    int d;
    d=f+g+h;
    cout<<d;
}

```



```

E:\Num1\Summer\PF\Lab Task
2
8
12
22
-----
Process exited after 9.711 seconds with return code 0.
Press any key to continue . . .

```

Lab 04

```

#include<iostream>

using namespace std;

void check_greator(float, float);

main(){
    int l, m;

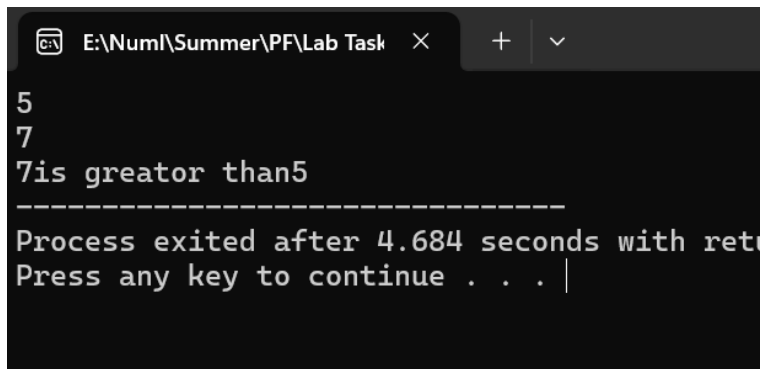
```

```

cin>>l>>m;
check_greator(l,m);
}

void check_greator(float p, float q){
if(p>q){
cout<<p<<"is greator than"<<q;
}
else{
cout<<q<<"is greator than"<<p;
}
}

```



```

5
7
7is greator than5
-----
Process exited after 4.684 seconds with ret
Press any key to continue . . . |

```

```

#include<iostream>

using namespace std;

print_square(int z){
z=z*z;
cout<<"Square is:"<<z;
}

```

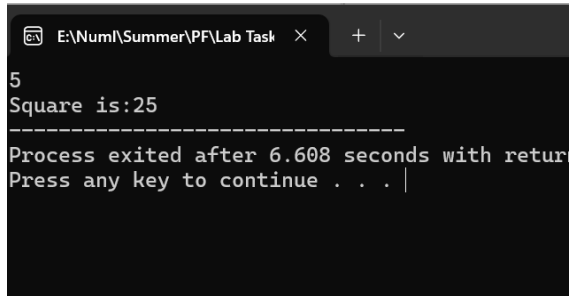
```

int main(){
    int s;

    cin>>s;

    print_square(s);
}

```



```

E:\Num1\Summer\PF\Lab Task
5
Square is:25
-----
Process exited after 6.608 seconds with return code 0
Press any key to continue . . .

```

```

#include<iostream>

using namespace std;

```

```

void print_last_digit(int);

```

```

int main(){
    int a,c;

    cin>>a;

    print_last_digit(a);
}

```

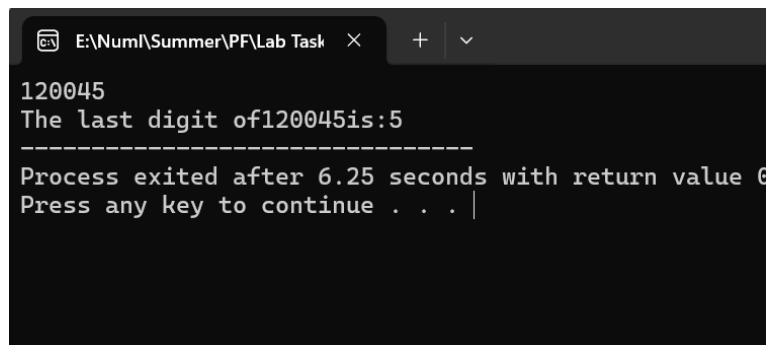
```

void print_last_digit(int a){
    int c;

    c=a%10;

    cout<<"The last digit of"<<a<<"is:"<<c;}

```



```
E:\Num1\Summer\PF\Lab Task
120045
The last digit of 120045 is: 5
-----
Process exited after 6.25 seconds with return value 0
Press any key to continue . . .
```

```
#include<iostream>
```

```
using namespace std;
```

```
float check_greator(float, float);
```

```
int last_digit(int);
```

```
void display_table(int);
```

```
main(){
```

```
int a,b;
```

```
cin>>a;
```

```
b=last_digit(a);
```

```
display_table(b);
```

```
}
```

```
int last_digit(int s){
```

```
int l;
```

```
l=s%10;
```

```
return l;
```

```
}
```

```
void display_table(int f){
```

```

int i;

for(int i=1; i<=10; i++){

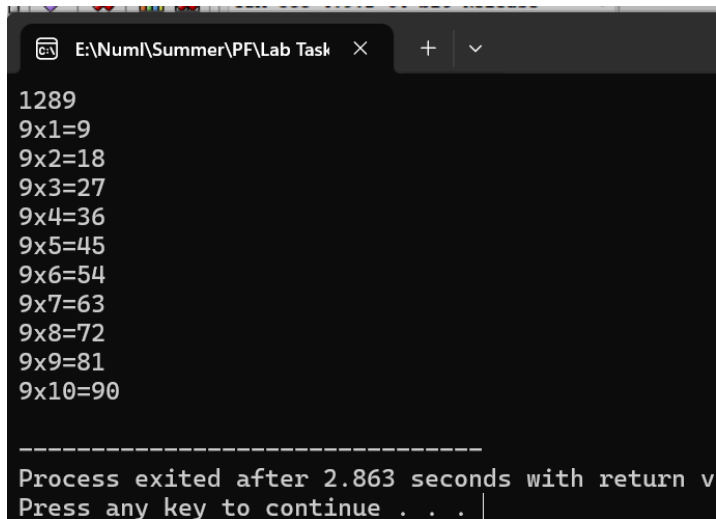
cout<<f<<"x"<<i<<"="<<f*i;

cout<<endl;

}

}

```



```

E:\Numl\Summer\PF\Lab Task
1289
9x1=9
9x2=18
9x3=27
9x4=36
9x5=45
9x6=54
9x7=63
9x8=72
9x9=81
9x10=90

-----
Process exited after 2.863 seconds with return v
Press any key to continue . . .

```

```

#include<iostream>

using namespace std;

int main(){

int r=3, c=3;

cout<<"Enter no rows:";

cin>>r;

cout<<"Enter no rows:";

cin>>c;

int arr[r][c];

for(int i=0; i<r; i++){

```

```

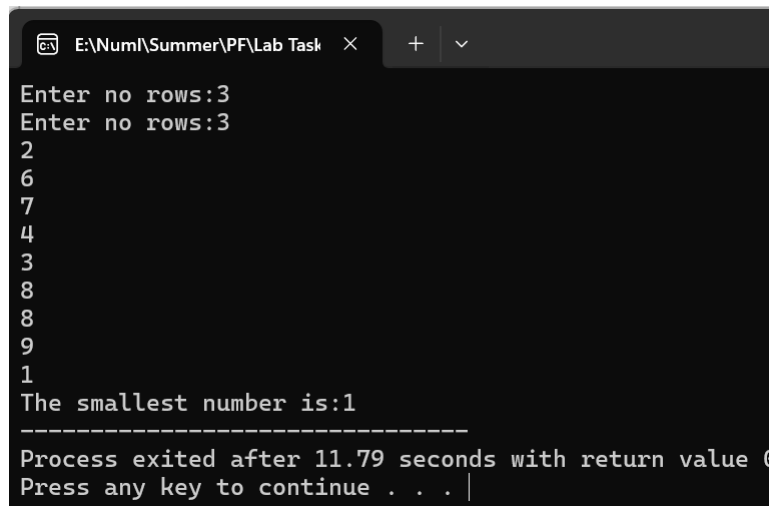
for(int j=0; j<c; j++){
cin>>arr[i][j];
}

int smallest=arr[0][0];
for(int i=0; i<r; i++){
for(int j=0; j<c; j++){

if(smallest>arr[i][j]){
smallest=arr[i][j];
}
}

}   cout<<"The smallest number is:"<<smallest;
}

```



```

E:\Numl\Summer\PF\Lab Task
Enter no rows:3
Enter no rows:3
2
6
7
4
3
8
8
9
1
The smallest number is:1
-----
Process exited after 11.79 seconds with return value 0
Press any key to continue . . . |

```



```
#include<iostream>

using namespace std;
```

```
void input(int &m1, int &m2, int &m3, int &m4, int &m5) {

    cout << "Enter bio Marks:" << endl;

    cin >> m1;

    cout << "Enter phs Marks:" << endl;

    cin >> m2;

    cout << "Enter chem Marks:" << endl;

    cin >> m3;

    cout << "Enter eng Marks:" << endl;

    cin >> m4;

    cout << "Enter math Marks:" << endl;

    cin >> m5;

}
```

```
void display_total_marks(int m1, int m2, int m3, int m4, int m5) {

    int total = 0;

    if (m1 > 49) total += m1;

    if (m2 > 49) total += m2;

    if (m3 > 49) total += m3;

    if (m4 > 49) total += m4;

    if (m5 > 49) total += m5;

    cout << "Total marks of subjects scoring above 49: " << total << endl;

}
```

```
int main() {

    int m1, m2, m3, m4, m5;
```

```

    input(m1, m2, m3, m4, m5);

    display_total_marks(m1, m2, m3, m4, m5);

    return 0;
}

```

```

E:\Numl\Summer\PF\Lab Task
Enter bio Marks:
70
Enter phs Marks:
87
Enter chem Marks:
99
Enter eng Marks:
45
Enter math Marks:
60
Total marks of subjects scoring above 49: 316
-----
Process exited after 11.62 seconds with return val

```

```

#include<iostream>

using namespace std;

void add(int a, int b){
    int sum=a+b;
    cout<<"Add:";
    cout<<sum;
    cout<<endl;
}

void subtract(int a, int b)
{

```

```
        int minus=a-b;
        cout<<"subtract:";
        cout<<minus;
        cout<<endl;
    }
```

```
void multiply(int a, int b)
{
```

```
    int mul=a*b;
    cout<<"multiply:";
    cout<<mul;
    cout<<endl;
}
```

```
void division(int a, int b){
```

```
    int divide=a/b;
    cout<<"Divide:";
    cout<<divide;
    cout<<endl;
}
```

```
void square(int a)
```

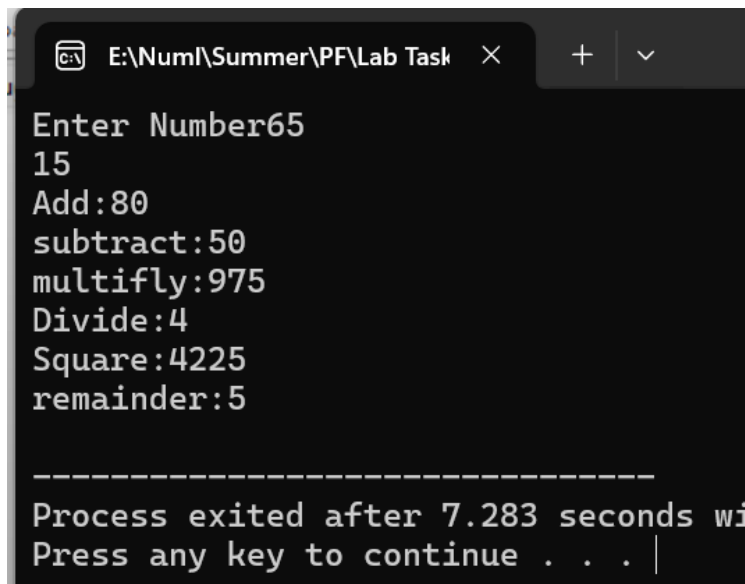
```
{
    int sqrt=a*a;
    cout<<"Square:";
    cout<<sqrt;
```

```
        cout<<endl;
    }
```

```
void rem(int a, int b){
```

```
    int rem=a%b;
    cout<<"remainder:";
    cout<<rem;
    cout<<endl;
}
```

```
int main(){
    int num1, num2;
    cout<<"Enter Number";
    cin>>num1>>num2;
    add(num1,num2);
    subtract(num1,num2);
    multify(num1,num2);
    division(num1,num2);
    square(num1);
    rem(num1,num2);
}
```



```
E:\Numl\Summer\PF\Lab Task
Enter Number65
15
Add:80
subtract:50
multifly:975
Divide:4
Square:4225
remainder:5

-----
Process exited after 7.283 seconds wi
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;

int main(){

int x=100;

cout<<x<<endl;

cout<<&x<<endl;

int *y=&x;

cout<<y<<endl;

cout<<&y<<endl;

cout<<*y<<endl;

*y=1000;

cout<<*y<<endl;

return 0;

}
```

```
E:\Numl\Summer\PF\Lab Task
100
0x6ffe1c
0x6ffe1c
0x6ffe10
100
1000

-----
Process exited after 9.551 seconds
Press any key to continue . . .
```

Lab 05

```
#include<iostream>

using namespace std;
```

```
void input();

void output(float []);
```

```
const int size=5;
```

```
int main(){

input();

}
```

```
void input(){

float a[size];

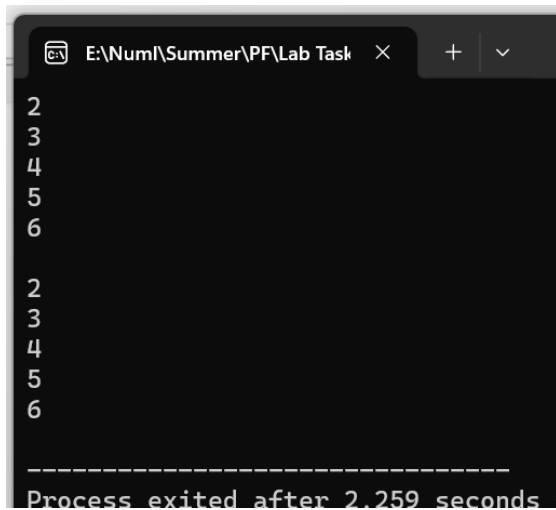
for(int i=0; i<size; i++){

cin>>a[i];

}
```

```
cout<<endl;
output(a);
}
```

```
void output(float b[]){
for(int i=0; i<size; i++){
cout<<b[i];
cout<<endl;
}
}
```

A screenshot of a terminal window with a dark background. The window title bar shows a file icon, the path "E:\Numl\Summer\PF\Lab Task", and standard window controls (close, maximize, minimize). The terminal displays the output of a program: the numbers 2, 3, 4, 5, and 6 are printed on five separate lines. This sequence is repeated once more. At the bottom of the terminal, a dashed line is followed by the text "Process exited after 2.259 seconds".

```
E:\Numl\Summer\PF\Lab Task  ×  +  ▾
2
3
4
5
6
2
3
4
5
6
-----
Process exited after 2.259 seconds
```

```
#include<iostream>

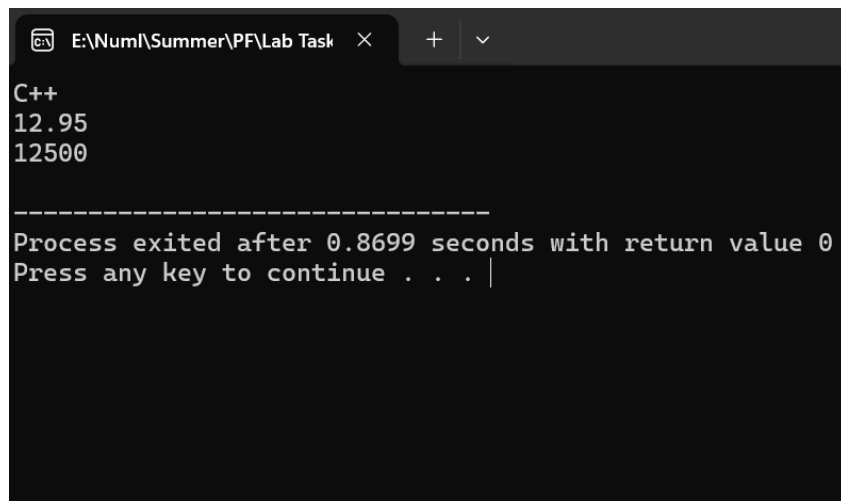
using namespace std;

struct book{
    string name;
    float price;
    int pages;
```

```
};
```

```
int main()
{
    book a;
    a.name="C++";
    a.price=12.95;
    a.pages=12500;

    cout<<a.name<<endl;
    cout<<a.price<<endl;
    cout<<a.pages<<endl;
}
```

A screenshot of a Windows command prompt window. The title bar shows the file path "E:\Num1\Summer\PF\Lab Task" and standard window controls. The terminal output shows the program's results: "C++", "12.95", and "12500" on separate lines. Below these, a separator line of dashes is followed by the message "Process exited after 0.8699 seconds with return value 0" and a prompt "Press any key to continue . . . |".

```
E:\Num1\Summer\PF\Lab Task  X  +  v
C++
12.95
12500

-----
Process exited after 0.8699 seconds with return value 0
Press any key to continue . . . |
```

```
#include<iostream>

using namespace std;
```



```
struct student{
```

```
    int roll;
```

```
    string name;
```

```
    float fee;
```

```
    int semester;
```

```
};
```

```
int main(){
```

```
    student a;
```

```
    cin>>a.roll;
```

```
    cin>>a.name;
```

```
    cin>>a.fee;
```

```
    cin>>a.semester;
```

```
    cout<<a.roll<<endl;
```

```
    cout<<a.name<<endl;
```

```
    cout<<a.fee<<endl;
```

```
    cout<<a.semester<<endl;
```

```
}
```

```
E:\Num\Summer\PF\Lab Task  ×  +  ▾  
4500  
ijlal  
90000  
4  
4500  
ijlal  
90000  
4  
  
-----  
Process exited after 42.78 seconds with return value 0  
Press any key to continue . . . |
```